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SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Configure a WPA2 Enterprise WLAN on the WLC

Addressing Table

Device	Interface	IP Address
R1	G0/0/0.5	192.168.5.1/24
	G0/0/0.200	192.168.200.1/24
	G0/0/1	172.31.1.1/24
SW1	VLAN 200	192.168.200.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.200.254/24
RADIUS/SNMP Server	NIC	172.31.1.254/24
Admin PC	NIC	192.168.200.200/24

Objectives

In this activity, you will configure a new WLAN on a wireless LAN controller (WLC), including the VLAN interface that it will use. You will configure the WLAN to use a RADIUS server and WPA2-Enterprise to authenticate users. You will also configure the WLC to use an SNMP server.

Configure a new VLAN interface on a WLC.

Configure a new WLAN on a WLC.

Configure a new scope on the WLC internal DHCP server.

Configure the WLC with SNMP settings.

Configure the WLC to use a RADIUS server to authenticate WLAN users.

Secure a WLAN with WPA2-Enterprise.

Connect hosts to the new WLC.

Background / Scenario

You have already configured and tested the WLC with an existing WLAN. You configured WPA2-PSK for that WLAN because it was to be used in a smaller business. You have been asked to configure and test a WLC topology that will be used in a larger enterprise. You know that WPA2-PSK does not scale well and is not appropriate to use in an enterprise network. This new topology will use a RADIUS server and WPA2-Enterprise to authenticate WLAN users. This allows administration of the user accounts from a central location and provides enhanced security and transparency because each account has its own username and password. In addition, user activity is logged on the server.

In this lab, you will create a new VLAN interface, use that interface to create a new WLAN, and secure that WLAN with WPA2-Enterprise. You will also configure the WLC to use the enterprise RADIUS server to authenticate users. In addition, you will configure the WLC to use a SNMP server.

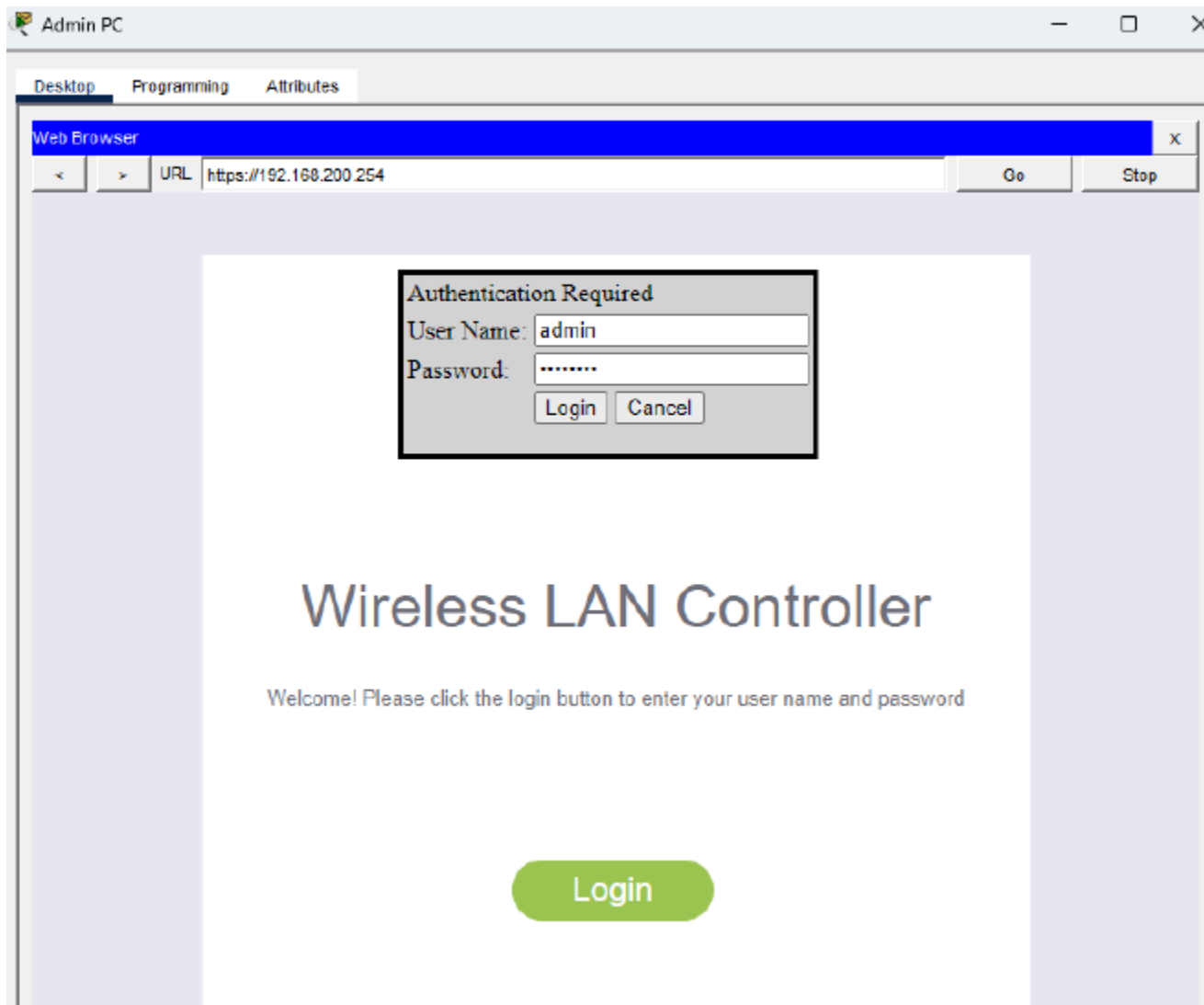
Instructions

Part 1: Create a new WLAN

Step 1: Create a new VLAN interface.

Each WLAN requires a virtual interface on the WLC. These interfaces are known as dynamic interfaces. The virtual interface is assigned a VLAN ID and traffic that uses the interface will be tagged as VLAN traffic. This is why connections between the APs, the WLC, and the router are over trunk ports. For the traffic from multiple WLANs to be transported through the network, traffic for the WLAN VLANs must be trunked.

- a. Open the browser from the desktop of Admin PC. Connect to the IP address of the WLC over HTTPS.
- b. Login with the username **admin** and password **Cisco123**.



- c. Click the **Controller** menu and then click **Interfaces** from the menu on the left. You will see the default virtual interface and the management interface to which you are connected.
- d. Click the **New** button in the upper right-hand corner of the page. You may need to scroll the page to the right to see it.
- e. Enter the name of the new interface. We will call it **WLAN-5**. Configure the VLAN ID as **5**. This is the VLAN that will carry traffic for the WLAN that we create later. Click **Apply**. This leads to a configuration screen for the VLAN interface.
- f. First, configure the interface to use physical port number **1**. Multiple VLAN interfaces can use the same physical port because the physical interfaces are like dedicated trunk ports.
- g. Address the interface as follows:
IP Address: **192.168.5.254**
Netmask: **255.255.255.0**
Gateway: **192.168.5.1**
Primary DHCP server: **192.168.5.1**

Packet Tracer - Configure a WPA2 Enterprise WLAN on the WLC

User traffic for the WLAN that uses this VLAN interface will be on the 192.168.5.0/24 network. The default gateway is the address of an interface on router R-1. A DHCP pool has been configured on the router. The address that we configure here for DHCP tells the WLC to forward all DHCP requests that it receives from hosts on the WLAN to the DHCP server on the router.

- h. Be sure to click **Apply** to enact your changes and click **OK** to respond to the warning message. Click **Save Configuration** so that your configuration will be in effect when the WLC restarts.

The screenshot shows the Admin PC interface for a Cisco Wireless LAN Controller (WLC). The browser window displays the URL `https://192.168.200.254/frameInterfaceEdit.html`. The interface has a top navigation bar with tabs for Desktop, Programming, and Attributes. Below this is a Cisco logo and a menu with options: MONITOR, WLANs, CONTROLLER, WIRELESS, SECURITY, MANAGEMENT, COMMANDS, HELP, and FEEDBACK. The main content area is titled "Controller" and "Interfaces > Edit". It features a left sidebar with a tree view of configuration options: General, Inventory, Interfaces (selected), Interface Groups, Multicast, Internal DHCP Server, Mobility Management, Ports, NTP, CDP, Tunneling, IPv6, mDNS, and Advanced. The main panel is divided into sections: General Information, Configuration, Physical Information, and Interface Address. The General Information section shows the Interface Name as "WLAN-5" and the MAC Address as "00:D0:BC:66:D5:D2". The Configuration section includes checkboxes for Guest Lan and Quarantine, a text field for Quarantine Vlan Id set to "0", and a text field for NAS-ID. The Physical Information section includes text fields for Port Number (set to "1"), Backup Port (set to "0"), Active Port (set to "1"), and a checkbox for Enable Dynamic AP Management. The Interface Address section includes text fields for VLAN Identifier (set to "5"), IP Address (set to "192.168.5.254"), Netmask (set to "255.255.255.0"), and Gateway (set to "192.168.5.1"). Buttons for "< BACK" and "Apply" are located at the top right of the configuration area.

General Information	
Interface Name	WLAN-5
MAC Address	00:D0:BC:66:D5:D2

Configuration	
Guest Lan	☐
Quarantine	☐
Quarantine Vlan Id	0
NAS-ID	

Physical Information	
Port Number	1
Backup Port	0
Active Port	1
Enable Dynamic AP Management	☐

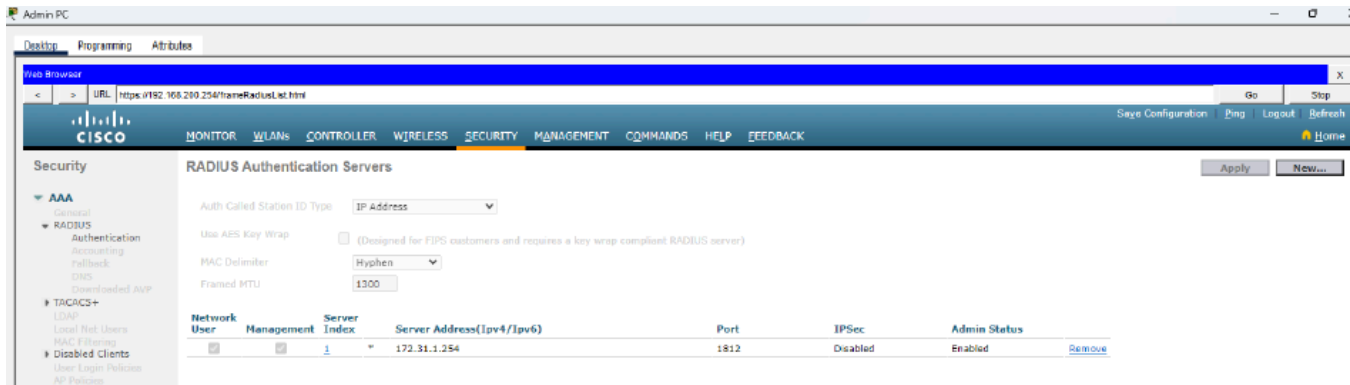
Interface Address	
VLAN Identifier	5
IP Address	192.168.5.254
Netmask	255.255.255.0
Gateway	192.168.5.1

Step 2: Configure the WLC to use a RADIUS server.

WPA2-Enterprise uses an external RADIUS server to authenticate WLAN users. Individual user accounts with unique usernames and passwords can be configured on the RADIUS server. Before the WLC can use the services of the RADIUS server, the WLC must be configured with the server address.

- Click the **Security** menu on the WLC.
- Click the **New** button and enter the IP address of the RADIUS server in the Server IP Address field.
- The RADIUS server will authenticate the WLC before it will allow the WLC to access the user account information that is on the server. This requires a shared secret value. Use **Cisco123**. Confirm the shared secret and click **Apply**.

Note: It is not a good practice to reuse passwords. This activity reuses passwords only to make the activity easier for you to complete and review.



Step 3: Create a new WLAN.

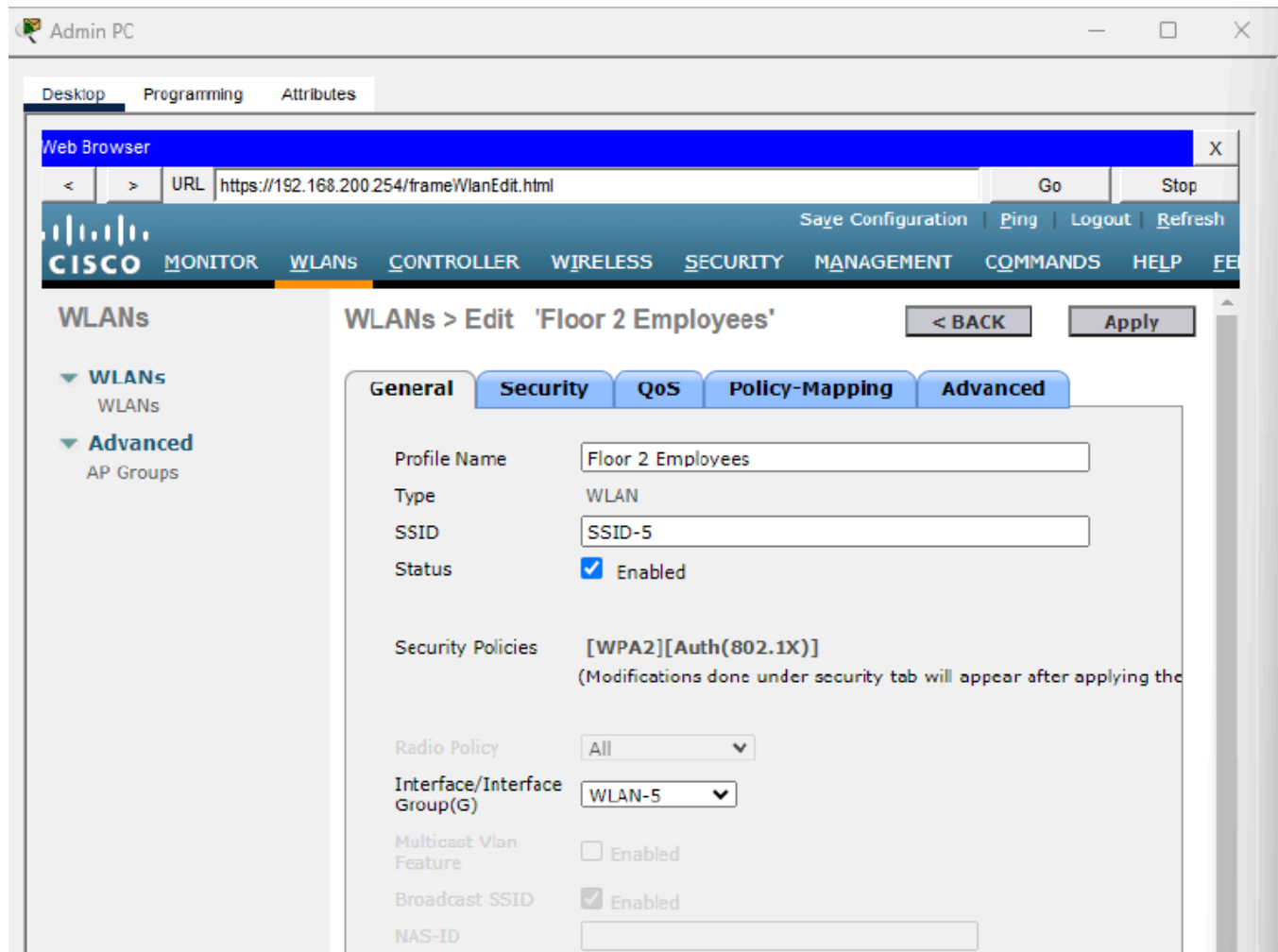
Create a New WLAN. Use the newly created VLAN interface for the new WLAN.

- Click the **WLANs** entry in the menu bar. Locate the dropdown box in the upper right-hand corner of the WLANs screen. It will say **Create New**. Click **Go** to create a new WLAN.
- Enter the **Profile Name** of the new WLAN. Use the profile name **Floor 2 Employees**. Assign an SSID of **SSID-5** to the WLAN. Change the ID drop down to **5**. Hosts will need to use this SSID to join the network. When you are done, click **Apply** to accept your settings.

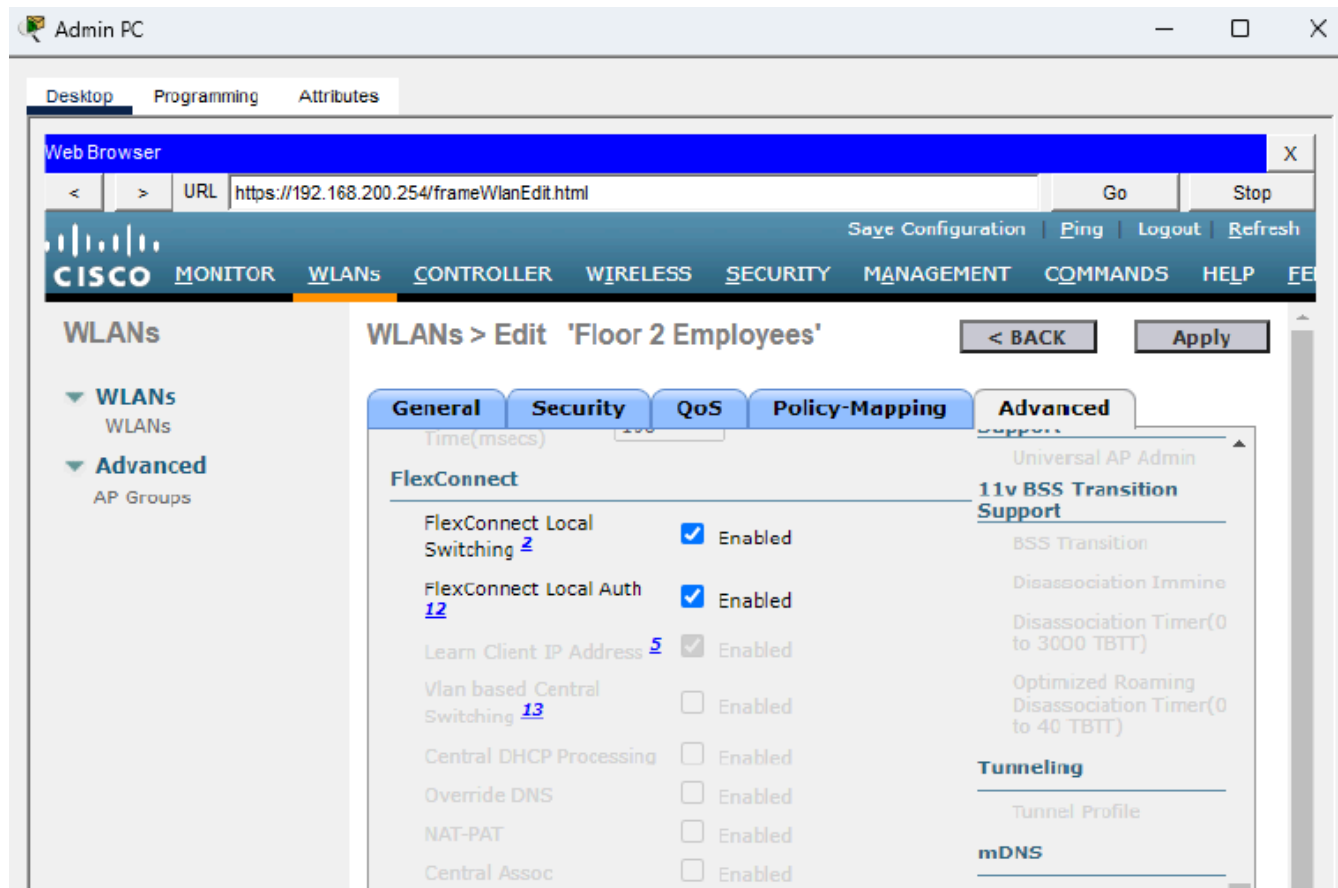
Note: The ID is an arbitrary value that is used as a label for the WLAN. In this case, we configured it as 5 to be consistent with VLAN for the WLAN. It could be any available value.

- Click **Apply** so that the settings go into effect.
- Now that the WLAN has been created you can configure features of the network. Click **Enabled** to make the WLAN functional. It is a common mistake to accidentally skip this step.
- Choose the VLAN interface that will be used for the new WLAN. The WLC will use this interface for user traffic on the network. Click the drop-down box for Interface/Interface Group (G). Select the interface that we created in Step 1.

Packet Tracer - Configure a WPA2 Enterprise WLAN on the WLC



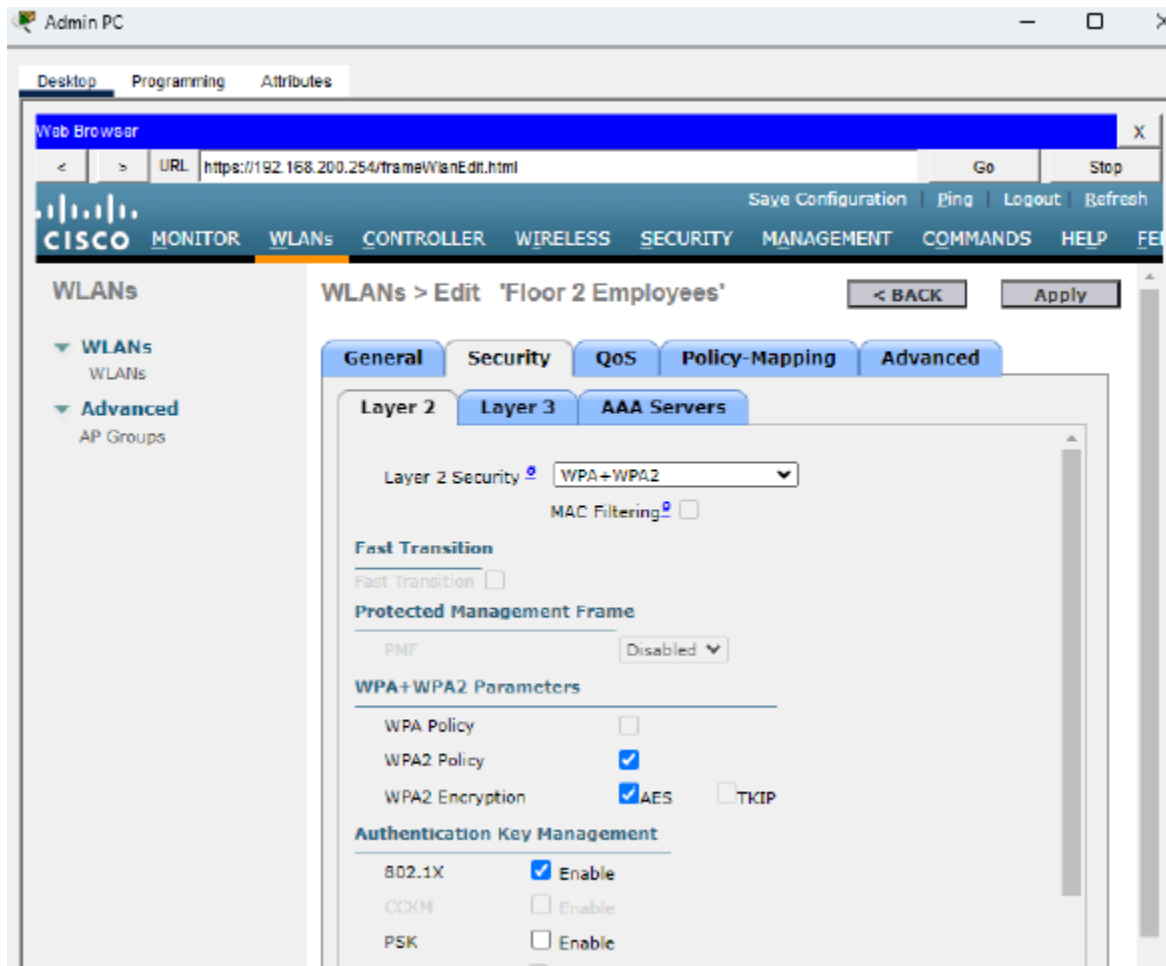
- f. Go to the Advanced tab. Scroll to **FlexConnect** section of the interface.
- g. Click to enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.
- h. Click **Apply** to enable the new WLAN. If you forget to do this, the WLAN will not operate.



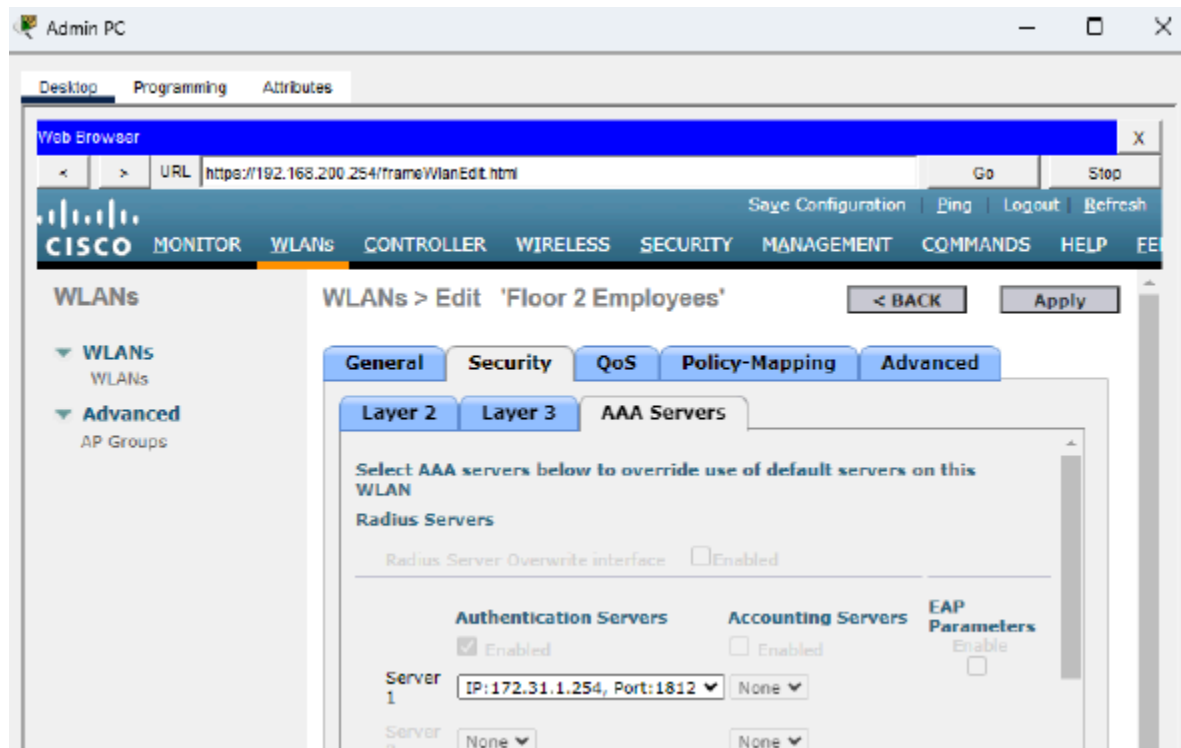
Step 4: Configure WLAN security.

Instead of WPA2-PSK, we will configure the new WLAN to use WPA2-Enterprise.

- Click the WLAN ID of the newly created WLAN to continue configuring it, if necessary.
- Click the Security tab. Under the Layer 2 tab, select **WPA+WPA2** from the drop-down box.
- Under WPA+WPA2 Parameters, enable **WPA2 Policy**. Click **802.1X** under Authentication Key Management. This tells the WLC to use the 802.1X protocol to authenticate users externally.



- d. Click the **AAA Servers** tab. Open the drop-down next to Server 1 in the Authentication Servers column and select the server that we configured in Step 2.
- e. Click **Apply** to enact this configuration. You have now configured the WLC to use the RADIUS server to authenticate users that attempt to connect to the WLAN.



Part 2: Configure a DHCP Scope and SNMP

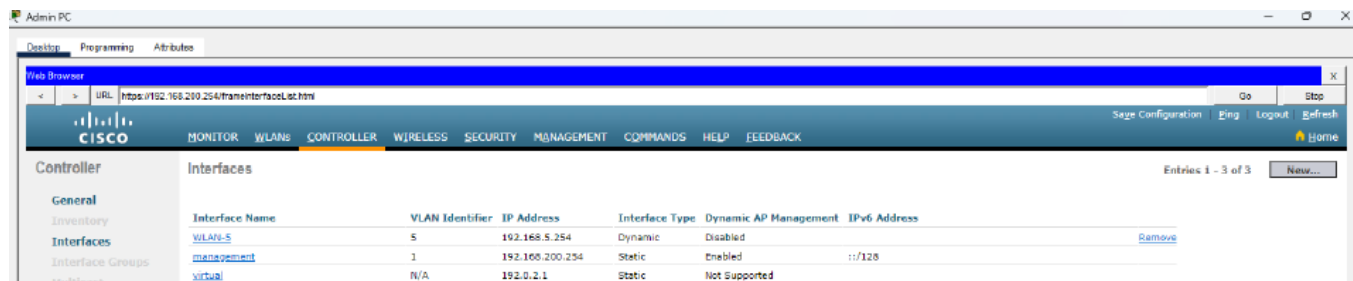
Step 1: Configure a DHCP Scope.

The WLC offers its own internal DHCP server. Cisco recommends that the WLAN DHCP server not be used for high-volume DHCP services, such as that required by larger user WLANs. However, in smaller networks, the DHCP server can be used to provide IP addresses to LAPs that are connected to the wired management network. In this step, we will configure a DHCP scope on the WLC and use it to address LAP-1.

- Should be connected to the WLC GUI from Admin PC.
- Click the **Controller** menu and then click **Interfaces**.

What interfaces are present?

WLAN-5, management, and virtual.



- Click the **management** Interface. Record its addressing information here.

IP address: **192.168.200.254**

Netmask: **255.255.255.0**

Gateway: **192.168.200.1**

Primary DHCP server: **0.0.0.0**

The screenshot shows the Cisco WLC Admin PC interface. The 'Web Browser' window displays the URL `https://192.168.200.254/frameInterfaceEdit.html`. The 'CONTROLLER' tab is selected, and the 'Internal DHCP Server' section is expanded in the left-hand menu. The 'Interface Address' section shows the following configuration:

Field	Value
Port Number	1
Backup Port	0
Active Port	1
Enable Dynamic AP Management	☒
VLAN Identifier	1
IP Address	192.168.200.254
Netmask	255.255.255.0
Gateway	192.168.200.1

The 'DHCP Information' section shows the following configuration:

Field	Value
Primary DHCP Server	0.0.0.0
Secondary DHCP Server	
DHCP Proxy Mode	Global
Enable DHCP Option 82	☐

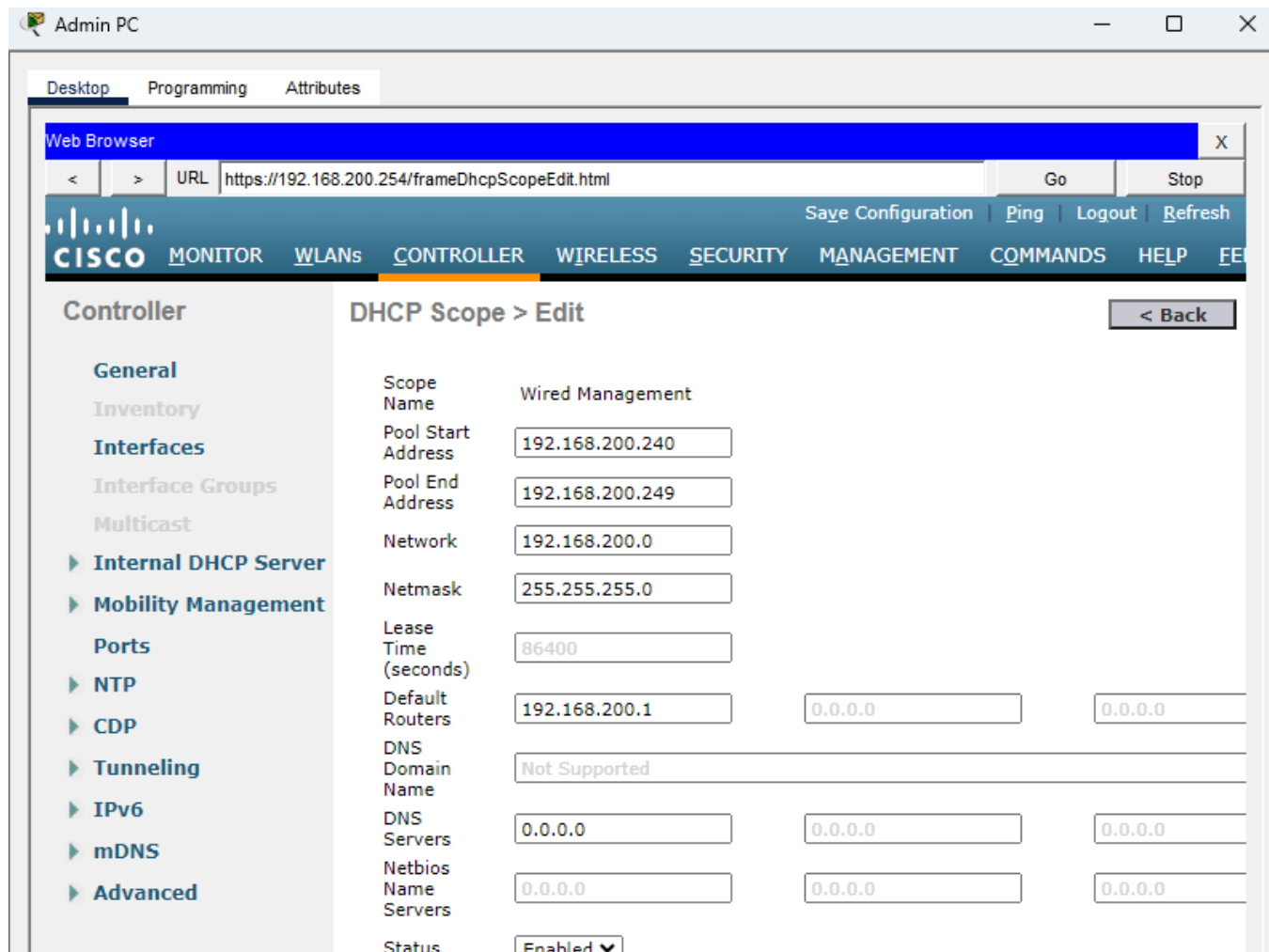
- We want the WLC to use its own DHCP server to provide addressing to devices on the wireless management network, such as lightweight APs. For this reason, enter the IP address of the WLC management interface as the primary DHCP server address. Click **Apply**. Click **OK** to acknowledge any warning messages that appear.
- In the left-hand menu, expand the **Internal DHCP Server** section. Click **DHCP Scope**.
- To create a DHCP scope, click the **New...** button.
- Name the scope **Wired Management**. You will configure this DHCP scope to provide addresses to the wired infrastructure network that connects the Admin PC, WLC-1, and LAP-1.
- Click **Apply** to create the new DHCP scope.
- Click the new scope in the DHCP Scopes table to configure addressing information for the scope. Enter the following information.
Pool Start Address: **192.168.200.240**
Pool End Address: **192.168.200.249**
Status: **Enabled**

Packet Tracer - Configure a WPA2 Enterprise WLAN on the WLC

Provide the values for **Network**, **Netmask**, and **Default Routers** from the information you gathered in Step 1c.

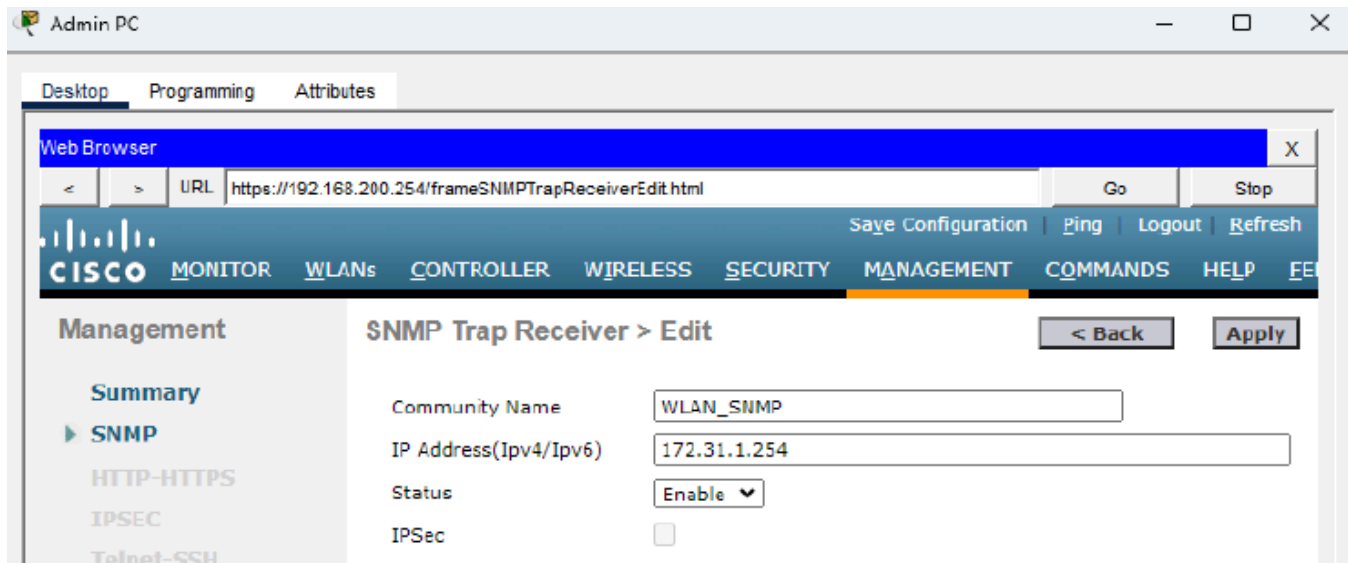
- j. Click **Apply** to activate the configuration. Click **Save Configuration** in the upper-right-hand corner of the WLC interface to save your work so that it is available when the WLC restarts.

The internal DHCP server will now provide an address to LAP-1 after a brief delay. When LAP-1 has its IP address, the CAPWAP tunnel will be established and LAP-1 will be able to provide access to the Floor 2 Employees (SSID-5) WLAN. If you move the mouse over LAP-1 in the topology, you should see its IP address, the status of the CAPWAP tunnel, and the WLAN that LAP-1 is providing access to.



Step 2: Configure SNMP

- a. Click the **Management** menu in the WLC GUI and expand the entry for **SNMP** in the left-hand menu.
- b. Click **Trap Receivers** and then **New...**
- c. Enter the community string as **WLAN_SNMP** and the IP address of the server at **172.31.1.254**.
- d. Click **Apply** to finish the configuration.



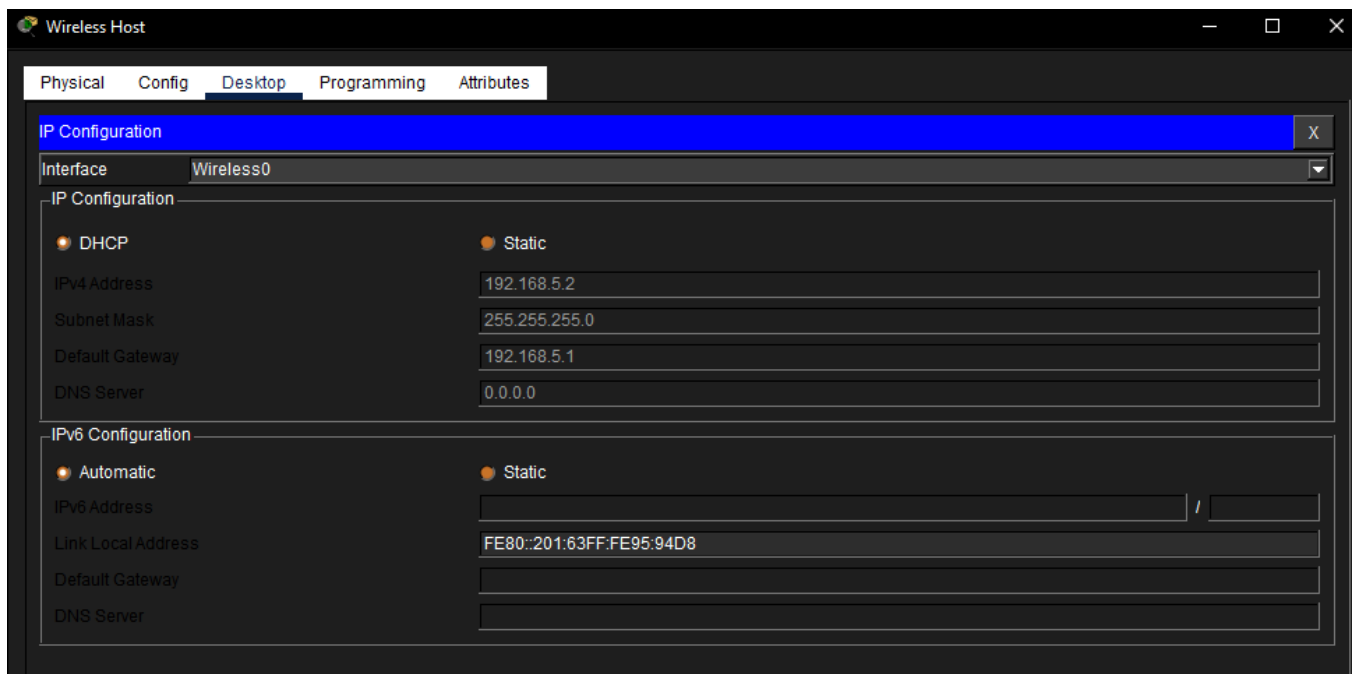
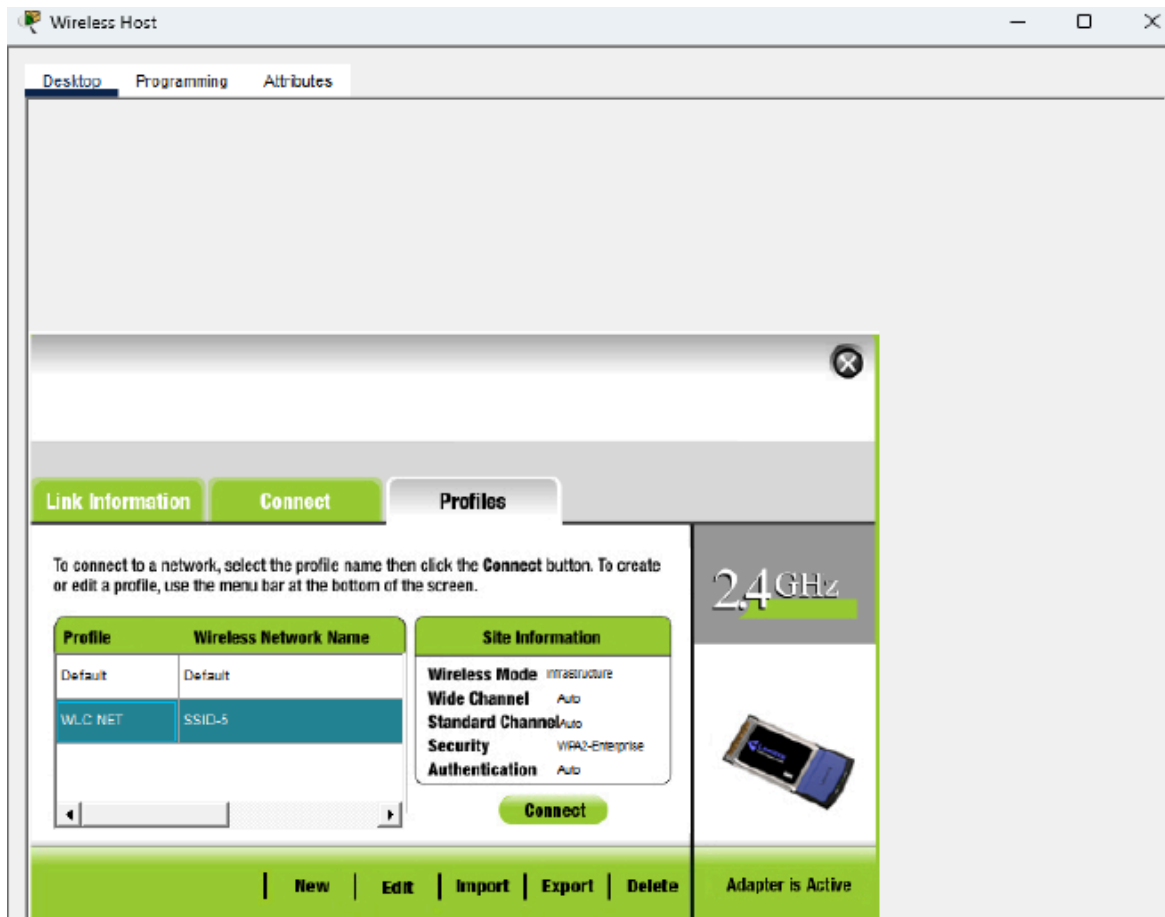
Part 3: Connect Hosts to the Network

Step 1: Configure a host to connect to the enterprise network.

In the Packet Tracer PC Wireless client app, you must configure a WLAN Profile in order to attach to a WPA2-Enterprise WLAN.

- a. Click Wireless Host and open the **PC Wireless** app.
- b. Click the **Profiles** tab and then click **New** to create a new profile. Name the profile **WLC NET**.
- c. Highlight the Wireless Network Name for the WLAN that we created earlier and click **Advanced Setup**.
- d. Verify that the SSID for the wireless LAN is present and then click **Next**. Wireless Host should see SSID-5. If it does not, move the mouse over LAP-1 to verify that it is communicating with the WLC. The popup box should indicate that LAP-1 is aware of SSID-5. If it is not, check the WLC configuration. You can also manually enter the SSID.
- e. Verify that the DHCP network setting is selected and click **Next**.
- f. In the Security drop down box, select **WPA2-Enterprise**. Click **Next**.
- g. Enter login name **user1** and the password **User1Pass** and click **Next**.
- h. Verify the Profile Settings and click **Save**.
- i. Select the **WLC NET** profile and click the **Connect to Network** button. After a brief delay, you should see the Wireless Host connect to LAP-1. You can click the Fast Forward Time button to speed up the process if it seems to be taking too long.
- j. Confirm that Wireless Host has connected to the WLAN. Wireless Host should receive an IP address from the DHCP server that is configured for hosts on R1. The address will be in the 192.168.5.0/24 network. You may need to click the Fast Forward Time button speed up the process.

Packet Tracer - Configure a WPA2 Enterprise WLAN on the WLC



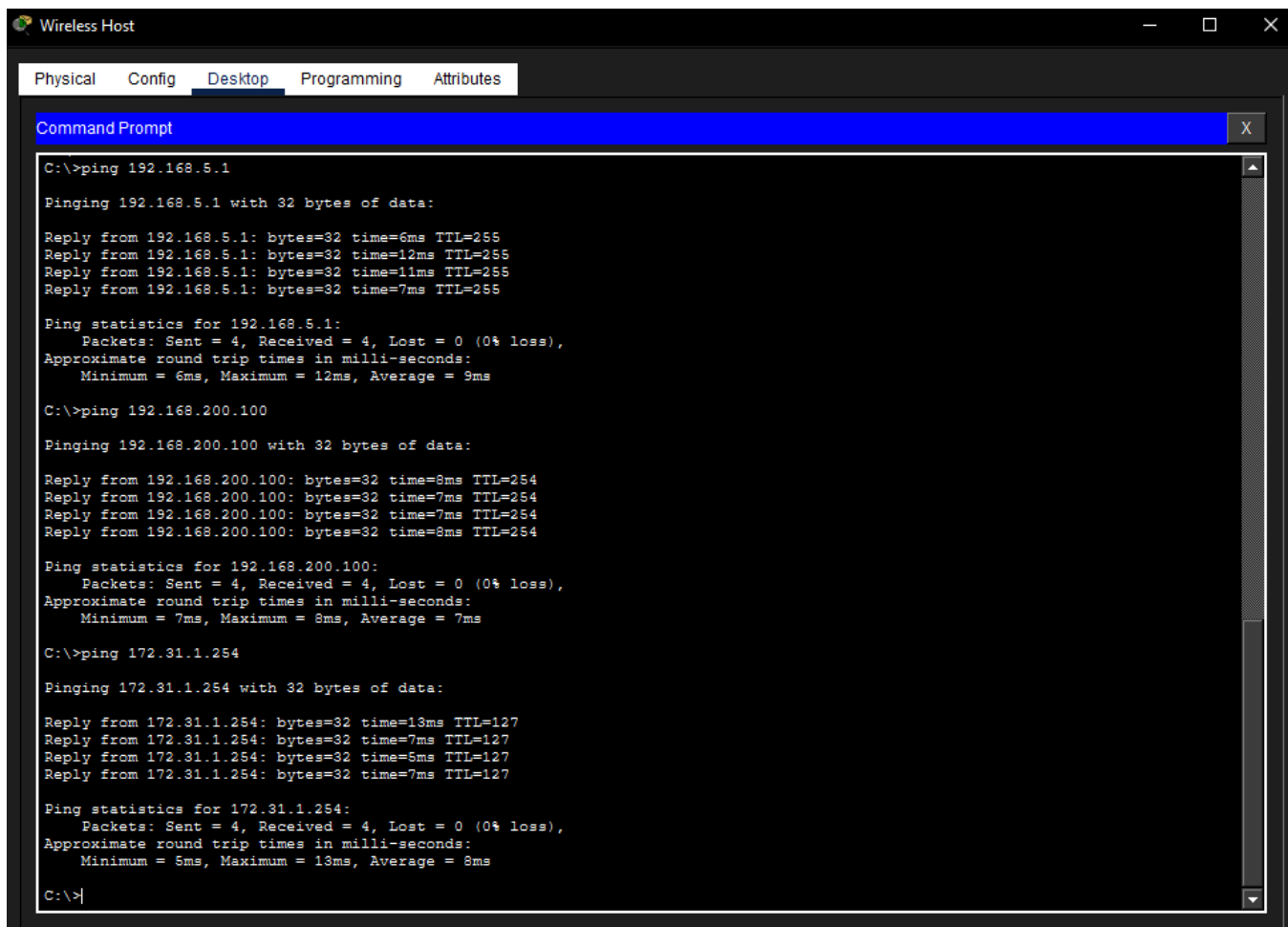
Step 2: Test Connectivity.

- a. Close the PC Wireless app.
- b. Open a command prompt and confirm that Wireless Host laptop has obtained an IP address from the WLAN network.

What network should the address be in? Explain.

The assigned IP address belongs to the 192.168.5.0/24 subnet, specifically yielding 192.168.5.2. This happens because the wireless host associates with the "Floor 2 Employees" wireless network (SSID-5), which is mapped directly to the VLAN 5 interface on the Wireless LAN Controller (WLC). Consequently, the laptop pulls its IP configuration via DHCP from a designated pool on Router R1, the default gateway managing routing for VLAN 5.

- c. Ping the default gateway, SW1, and the RADIUS server. Success indicates full connectivity within this topology.



```
Wireless Host
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.168.5.1

Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time=6ms TTL=255
Reply from 192.168.5.1: bytes=32 time=12ms TTL=255
Reply from 192.168.5.1: bytes=32 time=11ms TTL=255
Reply from 192.168.5.1: bytes=32 time=7ms TTL=255

Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 12ms, Average = 9ms

C:\>ping 192.168.200.100

Pinging 192.168.200.100 with 32 bytes of data:

Reply from 192.168.200.100: bytes=32 time=8ms TTL=254
Reply from 192.168.200.100: bytes=32 time=7ms TTL=254
Reply from 192.168.200.100: bytes=32 time=7ms TTL=254
Reply from 192.168.200.100: bytes=32 time=8ms TTL=254

Ping statistics for 192.168.200.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 8ms, Average = 7ms

C:\>ping 172.31.1.254

Pinging 172.31.1.254 with 32 bytes of data:

Reply from 172.31.1.254: bytes=32 time=13ms TTL=127
Reply from 172.31.1.254: bytes=32 time=7ms TTL=127
Reply from 172.31.1.254: bytes=32 time=5ms TTL=127
Reply from 172.31.1.254: bytes=32 time=7ms TTL=127

Ping statistics for 172.31.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 13ms, Average = 8ms

C:\>|
```

Reflection Questions

1. The RADIUS server uses a dual authentication mechanism. What two things are authenticated by the RADIUS server? Why do think this is necessary?

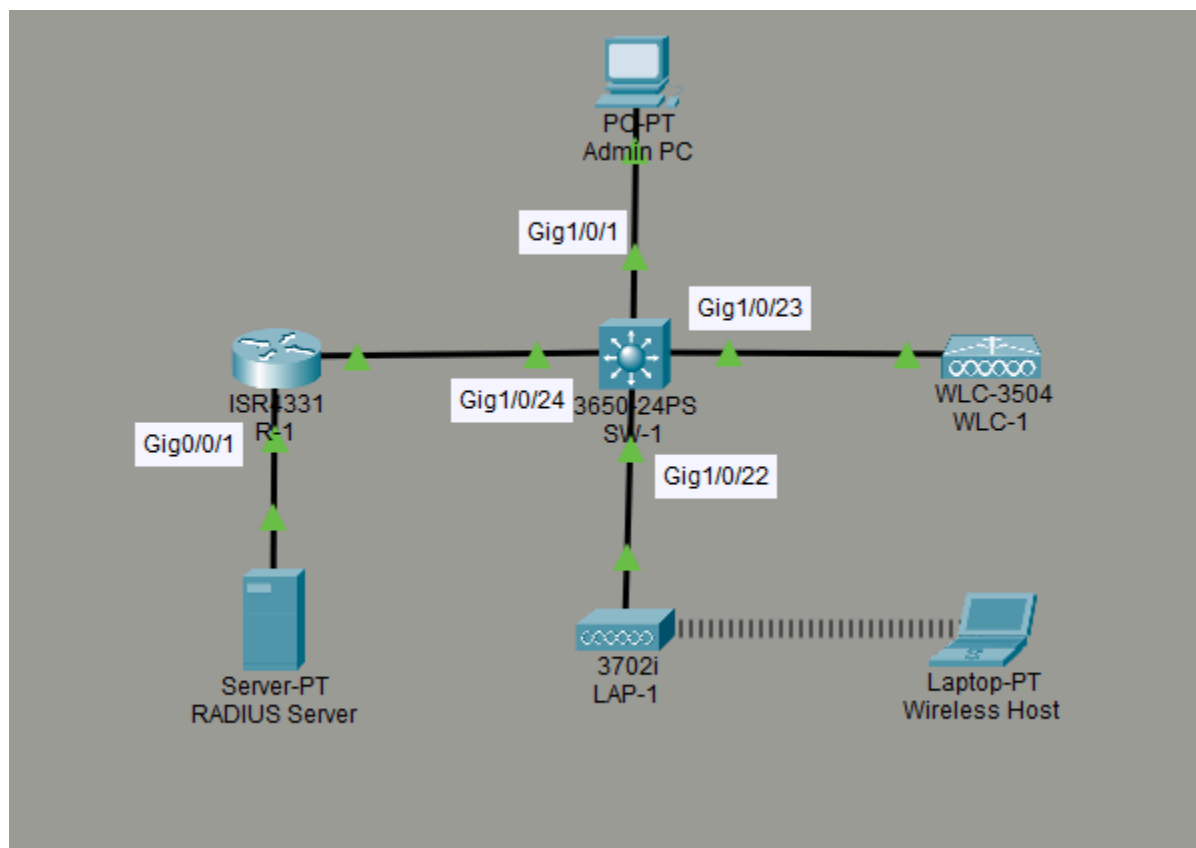
Authenticated Items: The RADIUS server validates both the Wireless LAN Controller (WLC) via a shared secret key and the individual end-user via unique login credentials (username/password).

Why it's necessary: This two-tier validation secures both the infrastructure and data access. Authenticating the WLC stops malicious actors from inserting unauthorized rogue access points or spoofed controllers into the network to intercept data. Authenticating the user ensures that only verified individuals are permitted on the network.

2. What are the advantages of WPA2-Enterprise over WPA2-PSK?

WPA2-Enterprise offers vastly superior scalability and network oversight compared to WPA2-PSK, which utilizes a single pre-shared key for every device. By implementing an 802.1X RADIUS setup, an enterprise gains centralized user management, individualized user tracking and logging, and the capability to instantly remove a single user's access without resetting the Wi-Fi password for everyone else. Additionally, it assigns unique, dynamic encryption keys to each user session.

Topology and Configurations



SW-1: enable configure terminal vlan 5 name WLAN-Traffic vlan 200 name Management exit interface GigabitEthernet1/0/1 switchport mode access switchport access vlan 200 no shutdown interface range GigabitEthernet1/0/22 – 24 switchport mode trunk no shutdown end write memory	R-1: enable configure terminal interface g0/0/0.5 encapsulation dot1Q 5 ip address 192.168.5.1 255.255.255.0 interface g0/0/0.200 encapsulation dot1Q 200 ip address 192.168.200.1 255.255.255.0 interface g0/0/0 no shutdown interface g0/0/1 ip address 172.31.1.1 255.255.255.0 no shutdown ip dhcp excluded-address 192.168.5.1 192.168.5.10 ip dhcp excluded-address 192.168.5.254 ip dhcp pool WLAN-HOSTS network 192.168.5.0 255.255.255.0 default-router 192.168.5.1 end write memory
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